THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The isotope used in the treatment of goitre is: 1

(a) cobalt-60 **(b)** carbon-14 **(c)** uranium-235 **(d)** iodine-131

2. Thomson's model of the atom is called the: 1

(a) quantum model **(b)** planetary model
(c) plum pudding model **(d)** shell model

3. The electronic configuration of sodium (Z = 11) is: 1

(a) 2, 9 **(b)** 2, 1 **(c)** 2, 8, 2 **(d)** 2, 8, 1

4. The isotope used as a fuel in nuclear reactors is: 1

(a) carbon-12 **(b)** iodine-131 **(c)** cobalt-60 **(d)** uranium-235

5. According to Bohr, electrons revolve in: 1

(a) straight lines **(b)** the nucleus
(c) fixed energy levels (shells) **(d)** random paths

6. The electronic configuration of chlorine (Z = 17) is: 1

(a) 2, 8, 7 **(b)** 2, 8, 8 **(c)** 2, 9, 6 **(d)** 2, 7, 8

7. The isotope used in the treatment of cancer is: 1

(a) chlorine-35 **(b)** iodine-131 **(c)** uranium-235 **(d)** cobalt-60

8. The mass of a proton is nearly equal to the mass of a: 1

(a) alpha particle only **(b)** neutron
(c) electron **(d)** photon

9. The valency of an element with the electronic configuration 2, 8, 2 is: 1

(a) 4 **(b)** 8 **(c)** 6 **(d)** 2

10. The three isotopes of hydrogen are: 1

(a) H, H2 and H3 **(b)** hydrogen, helium and lithium
(c) proton, neutron and electron **(d)** protium, deuterium and tritium

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is an atom electrically neutral? 2

12. Define atomic number and mass number. 2

13. What are isobars? Give an example. **2**

14. Write the relative charge and mass (in u) of electron, proton and neutron. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the number of protons, electrons and neutrons in $^{40}_{18}\mathrm{Ar}$ and $^{40}_{20}\mathrm{Ca}$. What are they called? **3**

16. What is valency? Give valency for Mg, Cl and Ne with reasons. **3**

17. An element has configuration 2, 8, 5. Name it, give its valency and the type of ion it can form. **3**

OR

Chlorine has two isotopes of mass 35 u and 37 u in the ratio 3 : 1. Calculate the average atomic mass. **3**

18. Calculate the mass number and the number of neutrons of an atom with atomic number 15 and 16 neutrons. **3**

OR

Why did Bohr's model say that atoms are stable in spite of electrons moving in orbits? **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between isotopes and isobars, giving two examples each. **5**

OR

Explain the Bohr-Bury scheme for the distribution of electrons in the shells with the rules and illustrate with potassium (Z = 19) and argon (Z = 18). **5**

20. An element X has atomic number 17 and mass number 35. Find p, n, e, the electronic configuration, the valency, and the formula of its compound with sodium. **5**

OR

Explain why an alpha particle (He nucleus) is deflected by a gold nucleus and why most alpha particles go undeviated, using the size of the nucleus compared with the atom. **5**

END OF QUESTION PAPER — Total: 40 Marks

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